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Planning for Eye Tracker Calibration

This article contains important guidelines and information to obtain proper eye tracking data.

1. Factors that negatively influence eye tracking

Room Light

Some external light sources can interfere with the eye tracker:

- Avoid fluorescent “Office lights” typically placed in the ceiling. They have an infrared component that can influence the trackers performance.
- Avoid sunlight from windows. Both because of the strong infrared component and because a window will provide changing light conditions over time.
- Create an environment with dispersed ambient light. For example point one or more incandescent lights towards a wall. Make sure there the level of light is adequate - too dark of lighting may make some respondents uncomfortable.
- If you have a respondent camera, make sure the light is bright enough to support the desired frame rate of the camera.

Human conditions

Some people can be difficult or even impossible to track due to some factors:

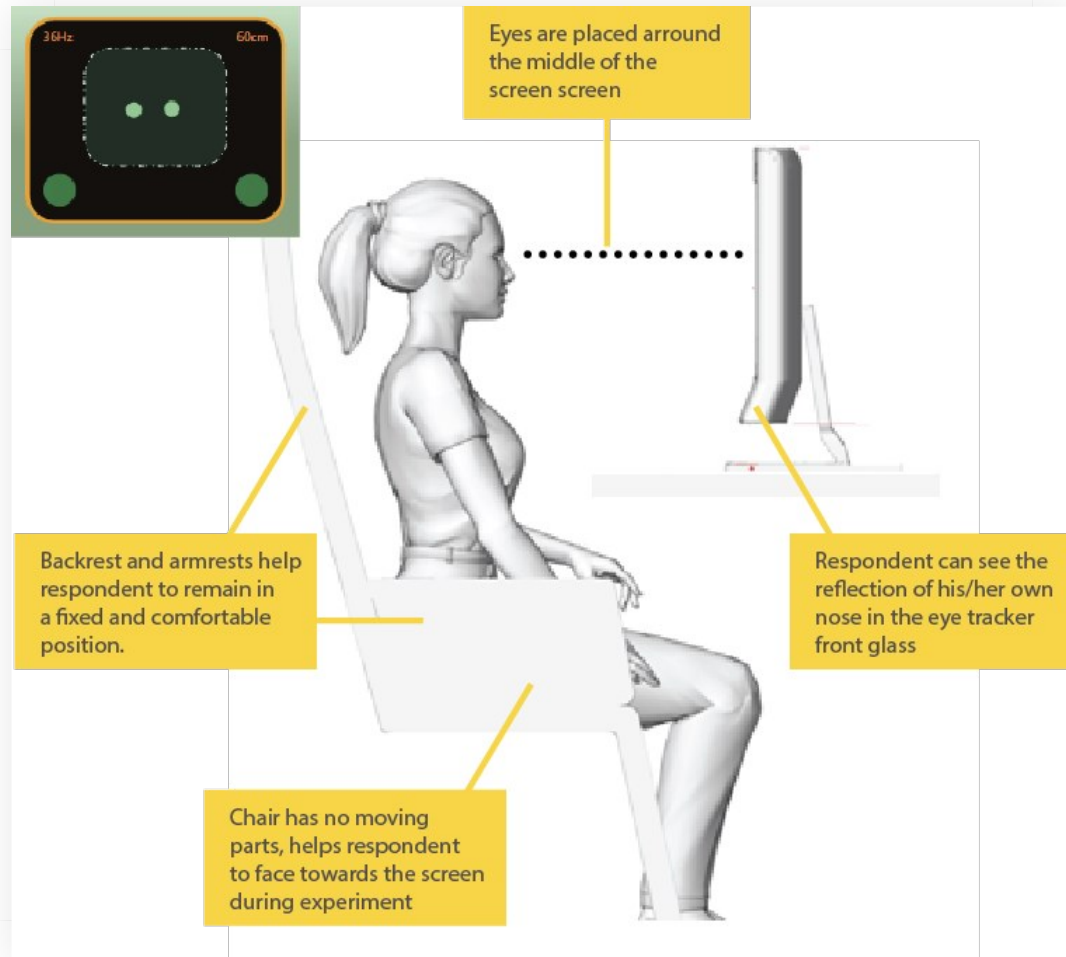
- Glasses - thick rim and/or severe near/far-sightedness
- Contact lenses - severe near/far-sightedness
- Astigmatism
- One-eyed
- Cross-eyed
- Previous Laser-eye surgery

People with glasses can be tracked but expect longer calibration time due to several calibration trials. If a subject has severe far or near-sightedness, it may be difficult to get a good calibration in any case.

2. Position Respondent

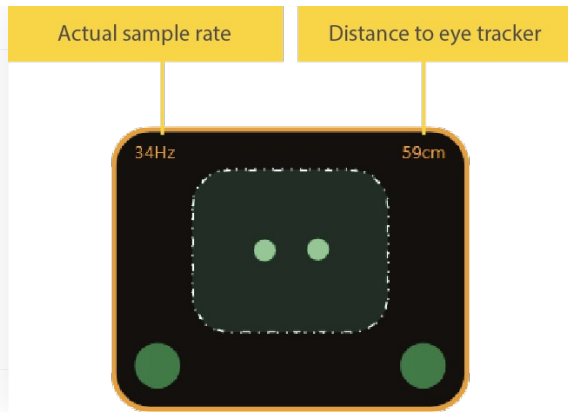
By using the iMotions's Eye Finder widget as a guide, make sure that

- The respondent's head is approximately 60 cm from the screen
- The respondent's eyes are approximately at the center of the screen.

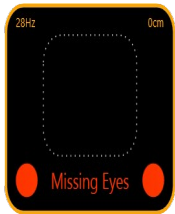
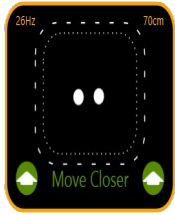


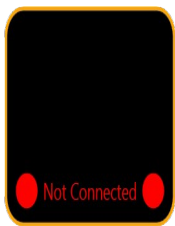
3. Eye Finder

The eye finder shows the position of the eyes according to the eye tracker, and the distance of the eyes to the eye tracker. Double click on the Eye Finder to detach it to a separate window.



Below is a list of all the eye finder states:

The eye finder cannot detect any eyes. Make sure that the respondent is positioned according to this guide. Also, make sure that the eye tracker is configured properly (according to the configuration guide of the specific eye tracker)	
The respondent is not positioned properly. Adjust the distance of the respondent from the eye tracker – according to the feedback presented in the eye finder	
The respondent is not positioned properly. Adjust the position of the respondent from the eye tracker – according to the feedback presented in the eye finder	

When the two dotted squares merge, and the distance indicator shows around 60 cm - the optimal positioning has been established	
The eye tracker is not connected. Make sure that the eye tracker is installed and configured properly (according to the installation and configuration guide of the specific eye tracker)	

4. Gaze Calibration

When you click the record button in iMotions, the eye tracking calibration will start automatically, after an instruction slide.

NOTE: This will only happen if the study and respondent is set to "Compact" and "One Screen Calibration" see [the guide here](#).

On the next screen you will see a shape moving across the screen.

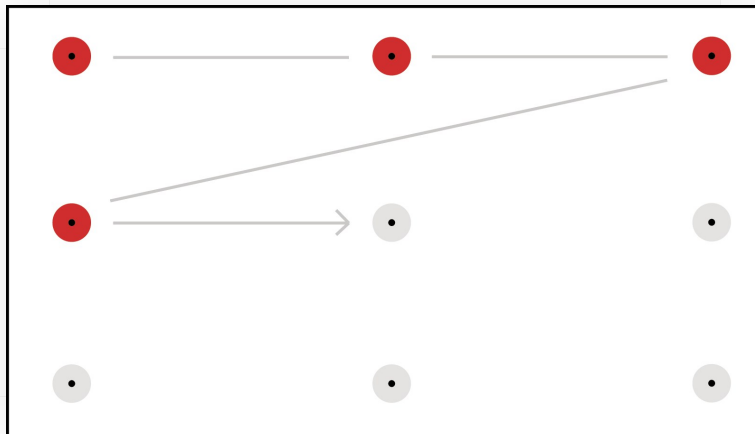
Follow the shape with your eyes and try to focus on the middle of it.

It is also possible to input a custom instruction slide for calibration. Go to "Preferences" select "Global Settings", and look at the bottom of "Calibration" tab. Here you can adjust the slide settings for "Compact Calibration". Select "Custom" to import a specific calibration instruction slide.

Compact Calibration

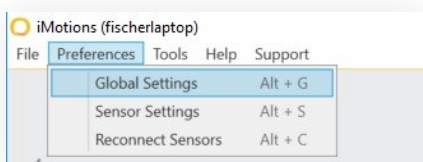
Calibration Instructions	☐ Default	☒ Custom
Instructions Position	☐ First	☒ After Sensors
Play Instructions On Key Press	☒ Yes	☐ No
Instructions Image/Video	Browse	
Instructions Timeout (s)	5	

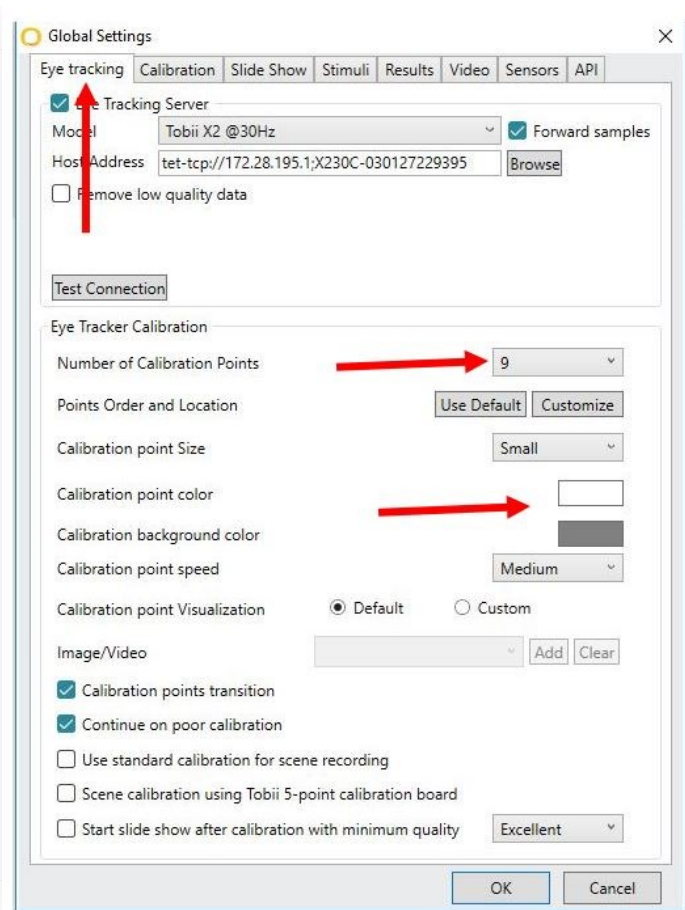
NOTE: Make it a habit to also tell the respondent, to follow the red circle with his eyes.



NOTE: To obtain the best results, ask the respondent to focus on the black dot in the middle of the red circle!

You can change the number of calibration points, and the size, color and speed of the calibration points under *Preferences* --> *Global Settings* and under *Calibration* tab, scroll down to *Eye Tracker Calibration*:





NOTE: For better eye tracking quality, use at least 9 calibration points.

When the respondent has finished the eye tracking calibration, you see the result which can either be Excellent, Good or Poor.

Excellent Calibration

Continue to test

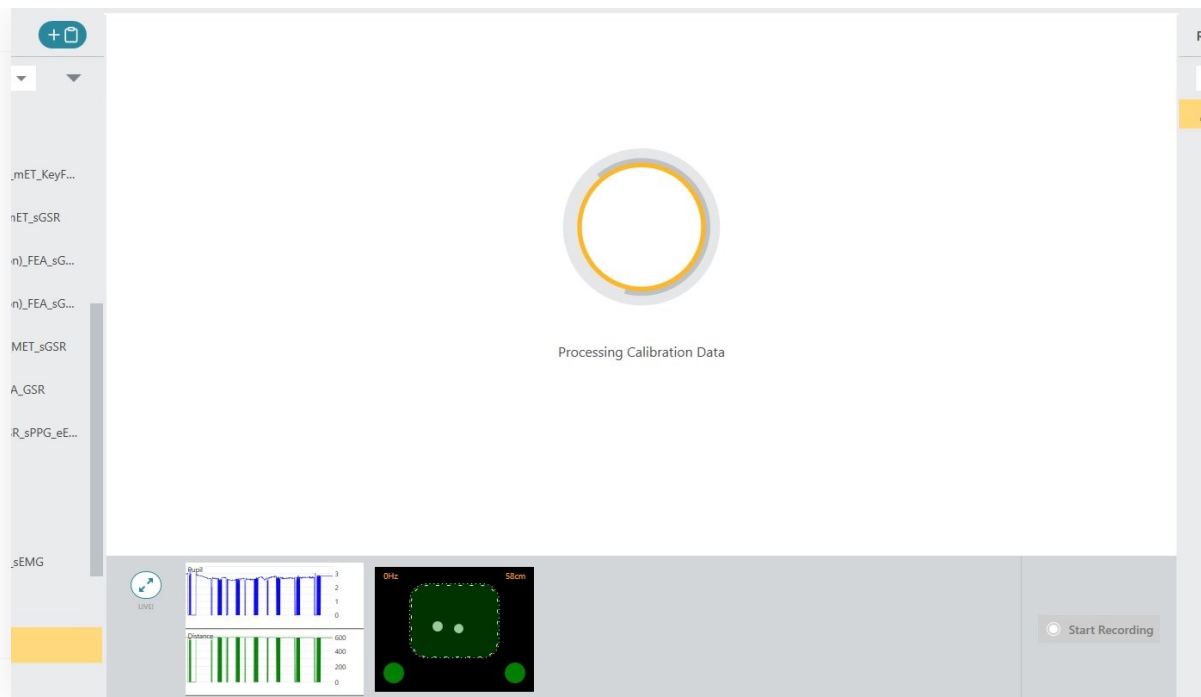
Good Calibration

Try to recalibrate once or twice to see if you can improve the calibration to become Excellent. Otherwise, continue to test

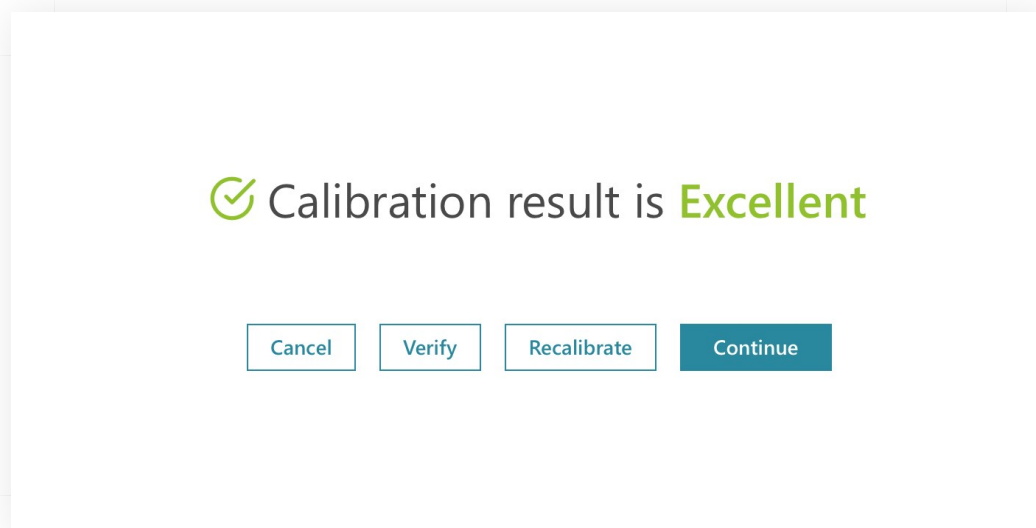
Poor Calibration

Try to recalibrate once or twice to see if you can improve the calibration to become Good or Excellent. If calibration never goes above poor you may want to skip the respondent altogether.

While you are waiting for the calibration to finish you will see a spinning wheel



Depending on the results a different scene will be shown. The optimal case will be Excellent, in which case you will see this:



If you did not get Excellent calibration, consider recalibration. The screen will show the following in the case of a good calibration:

✓ Calibration result is **Good**

You can select *Verify* to evaluate the calibration before continuing

Cancel

Verify

Recalibrate

Continue

In the case of a poor calibration, the screen will show the following, including tips on how to improve the calibration results:

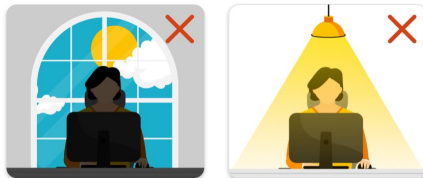
⚠ Calibration result is **Poor**

[View Help Center](#)

Check the following, and then try recalibrating

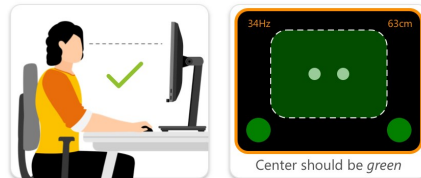
1 Room lighting conditions

Make sure that there is no bright light behind the respondent. Avoid fluorescent ceiling lights.



2 Respondent positioning

Make sure respondent is sitting at the correct distance from the eye tracker. Check the eye tracker widget status.



3 Screen angle

Make sure the screen is tilted so that the respondent can see the reflection of their eyes in the eyetracker



4 Respondent eyesight

Check with the respondent that they do not have eye problems such as:

- Cross-eyed
- Laser eye surgery
- Severe near- or far-sightedness
- Thick glasses
- Astigmatism

Cancel

Verify

Continue

Recalibrate

